SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Steffi. J (192524480) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

level - 3

2. Formulate the 8-puzzle problem (a 3x3 sliding tile puzzle) as a search problem by specifying the state representation, initial state, goal test, and the set of possible actions. Explain why the branching factor of this problem is at most 4.

3. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Analytical Problem Solving

1) Hospital AI Agent - Performance Measure and Rationality

Performance Measure

A performance measure evaluates how well the AI agent performs triage. For hospital system, it should include:

→ Accuracy of triage
→ Response time
→ Patient safety
→ Resource utilization
→ Patient outcome

Example: A good performance measure prioritizes saving lives while minimizing waiting time.

Concept of Rationality

An agent is rational if it selects actions that maximize expected performance based on:

→ Available Information
→ Past knowledge
→ Current situation.

Agent Rational

→ Prioritizes critical patients first
→ uses all available patient data correctly
→ Adapts to emergency situations
→ Minimizes risk

Q) Robot in Unknown Cave - Online/contigency Search

Concept

The Robot has:

→ No prior map

→ Limited sensor range

How it works

→ Robot senses nearby area

→ Chooses next move

→ updates internal map

→ Repeats untill exit is found

Map building

→ Stores Visited locations

→ Avoids revisiting paths

→ Gradually constructs full cave map

Risks of irreversible actions

Irreversible actions = Actions that cannot be undone

Examples:

→ Falling into a pit

→ Blocking path permenantly

Risks

→ Robot may be stuck

→ Cannot back track

Solution

→ Explore catiously

→ Prefer safe, reversible moves.

8- Puzzle Problem Formulation

State Representation

A 3X3 grid with tiles numbered 1-8 and one blank (0)

Example:

1	2	3
4	5	6
7	8	9

Initial State

Any random arrangement of tiles

Goal State

Tiles arranged in order:

1	2	3
4	5	6
7	8	9

Actions

Possible moves of the black tile

→ Move up

→ Move down

→ Move left

→ Move right

Branching Factor (why ≤4?)

→ Maximum moves from any state = 4

→ But depends on blank position:

Position	Moves
Corner	2
Edge	3
Center	4

So, Maximum branching factor = 4